Lecture Five

5.1 Introduction

Welcome to the fifth lecture on system development. In this lecture we shall discuss the techniques in information systems development.

5.2 Specific objectives
By the end of this lecture you should be able to

- Identify different techniques used in system development
- Relate the different techniques to its related methods
- Understand the different notations used in the different techniques

5.3 Lecture Outline

5.3.1 Introduction

5.3.2. Holistic Techniques

5.3.3 Data Techniques

5.3.4 Process Techniques

5.3.5 Object Oriented Techniques

5.3.6 Project Management Techniques

5.3.1 Introduction

There are various types of techniques that may be used during system development. These techniques can be classified into different categories. Each technique uses its own notations and presents a different view or perspective of that system.

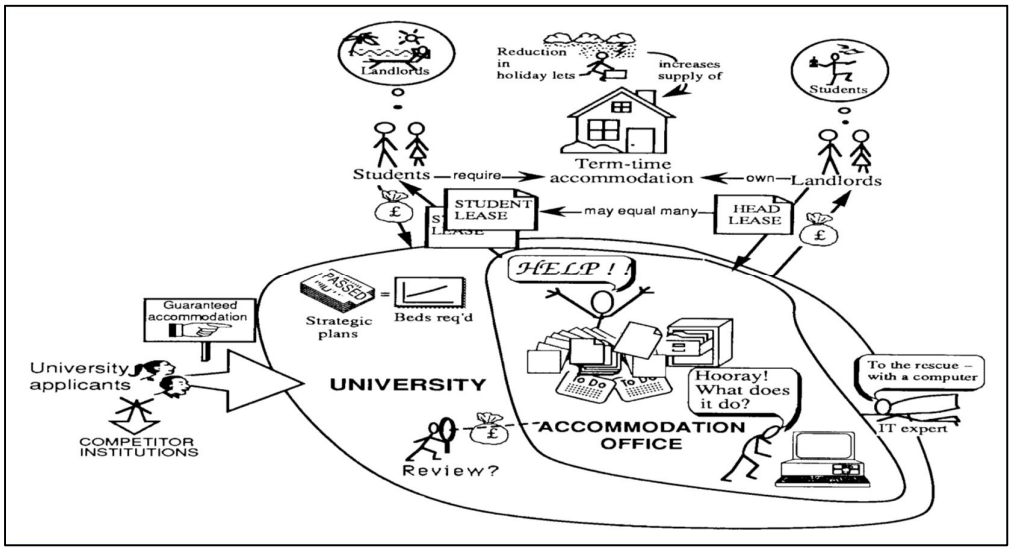
5.3.2. Holistic Techniques

Soft Systems Methodology (SSM for short) was developed by Peter Checkland and colleagues at the University of Lancaster. It is based upon systems theory, which provides an antidote to conventional, 'reductionist' scientific enquiry – with its tendency to 'reduce' phenomena into smaller and smaller components in order to study and understand them. Systems theory attempts to study the whole picture; the relation of component parts to each other, and to the wider picture - it is 'holistic.' SSM helps formulate and structure thinking about problems in complex, human situations. Its core is the construction of conceptual models and the comparison of those models with the real world. This process can greatly clarify those multifaceted problems with many

conflicting potential solutions, or no obvious way forward. It uses rich pictures , conceptual models and root definitions.

1. Rich Pictures

- Help to understand the problem situation in general at beginning of a project
- It is a pictorial caricature of an organization
- It is a tool for helping to explain what the organization is about
- It helps people to visualize and discuss their roles
- It can show the worries of individuals, potential conflict, and political issues



2.Root Definitions

This a short textual definition of the aims and means of the system to be modelled. It can be used to define two things that are otherwise both vague and difficult: problems and systems**.** A concise verbal description of the system which captures its essential nature of *Who* is doing *what* for *whom*, and to whom are they *answerable*, what *assumptions* are being made, and in what *environment* is this happening? or "A system to do **X**, by (means of) **Y**, in order to **Z"**
We identify this description using C(lient) A(ctor) T(ransformation) W(eltanschuung) O(wner) E(nvironment): CATWOE

Customers the victims or beneficiaries of T

Actors those who do T

Transformation process input into output

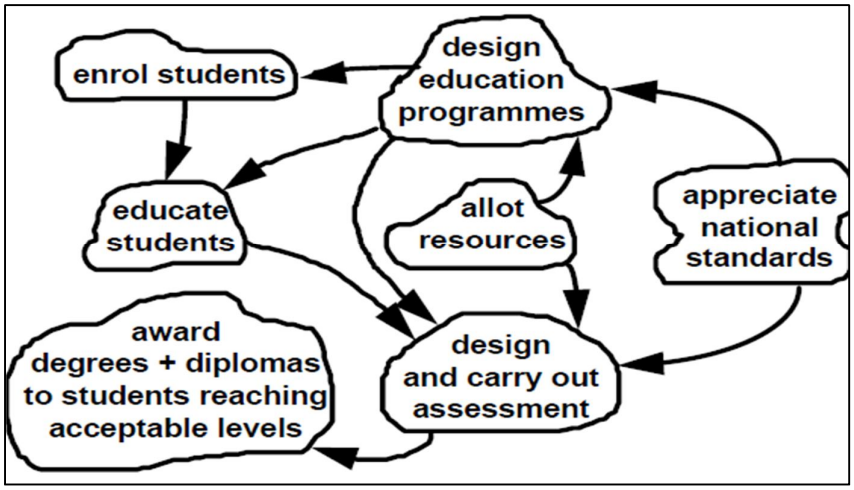
Weltanschauung the worldview that makes the T meaningful in context

Owners those with the power to stop T

Environmental constraints elements outside the system which are taken as given,but nevertheless affect its behavior.

5.3.2.2 Conceptual Models

Conceptual models demonstrate potential activities and their logical dependencies. The activities, which must be expressed in a verb noun phrase are placed in rough, hand drawn bubbles. The bubbles may be joined by arrows indicating dependence: - that one activity is consequent upon another - it cannot be performed, unless the other has been performed, or that it will be done poorly if the other is done poorly. activities usually makes for a reasonably understandable model. If more detail or complexity is required, then the system may be modelled at a higher level of resolution. (This is equivalent to levelling in a data flow diagram). Any activity in a conceptual model may be taken to represent a system in its own right.



5.3.2.4 Cognitive Mapping

This is a model of the system concepts used to communicate the nature of a problem and the concepts are related to others through an action orientation. The maps show short statements relating to the problem situation linked by arrows showing how one idea might lead to or have implications for another. It is seen as a formal modeling technique with rules for its development.

5.3.3 Data Techniques

5.3.3.1 Data-driven modelling

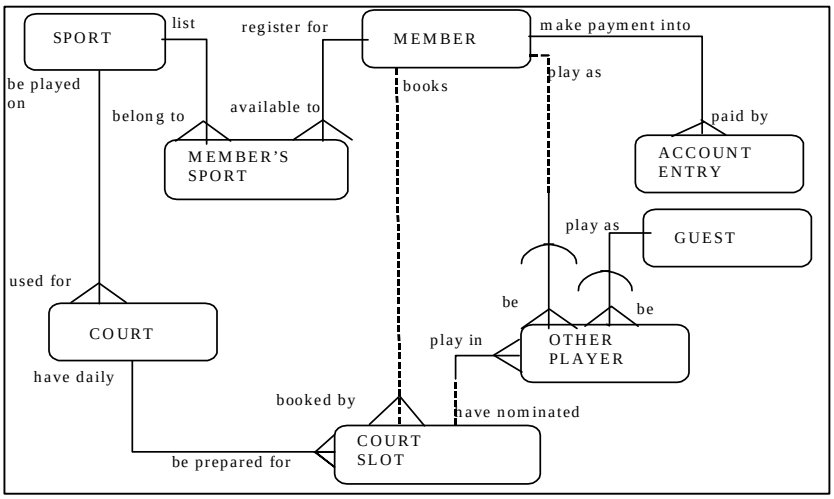
Many business systems are data-processing systems that are primarily driven by data. They are controlled by the data input to the system, with relatively little external event processing. Data-driven models show the sequence of actions involved in processing input data and generating an associated output. They are particularly useful during the analysis of requirements as they can be used to show end-to-end processing in a system.

1. Entity Modelling

An entity-relationship diagram (ERD) is a graphical representation of an information system that shows the relationship between people, objects, places, concepts or events within that system. An ERD is a data modeling technique that can help define business processes and can be used as the foundation for a relational database.

While useful for organizing data that can be represented by a relational structure, an entity-relationship diagram can't sufficiently represent semi-structured or unstructured data, and an ERD is unlikely to be helpful on its own in integrating data into a pre-existing information system.

Three main components of an ERD are the entities, which are objects or concepts that can have data stored about them, the relationship between those entities, and the cardinality, which defines that relationship in terms of numbers. For example, an ER diagram representing the information system for a company's sales department might start with graphical representations of entities such as the sales representative, the customer, the customer's address, the customer's order, the product and the warehouse. Then lines or other symbols can be used to represent the relationship between entities, and text can be used to label the relationships.



2. Normalisation

Database Normalisation is a technique of organizing the data in the database. Normalization is a systematic approach of decomposing tables to eliminate data redundancy and undesirable characteristics like Insertion, Update and Deletion. It is a multi-step process that puts data into tabular form by removing duplicated data from the relation tables. Normalization is used for mainly two purpose.

1. Eliminating redundant data.
2. Ensuring data dependencies make sense i.e data is logically stored.

Normalization rule are divided into following normal form.

1. First Normal Form
2. Second Normal Form
3. Third Normal Form

First Normal Form, no two Rows of data must contain repeating group of information i.e each set of column must have a unique value, such that multiple columns cannot be used to fetch the same row. Each table should be organized into rows, and each row should have a primary key that distinguishes it as unique.

The Primary key is usually a single column, but sometimes more than one column can be combined to create a single primary key. For example consider a table which is not in First normal form

Second Normal Form there must not be any partial dependency of any column on primary key. It means that for a table that has concatenated primary key, each column in the table that is not part of the primary key must depend upon the entire concatenated key for its existence. If any column depends only on one part of the concatenated key, then the table fails Second normal form.

Third Normal form applies that every non-prime attribute of table must be dependent on primary key, or we can say that, there should not be the case that a non-prime attribute is determined by another non-prime attribute. So this *transitive functional dependency* should be removed from the table and also the table must be in Second Normal form

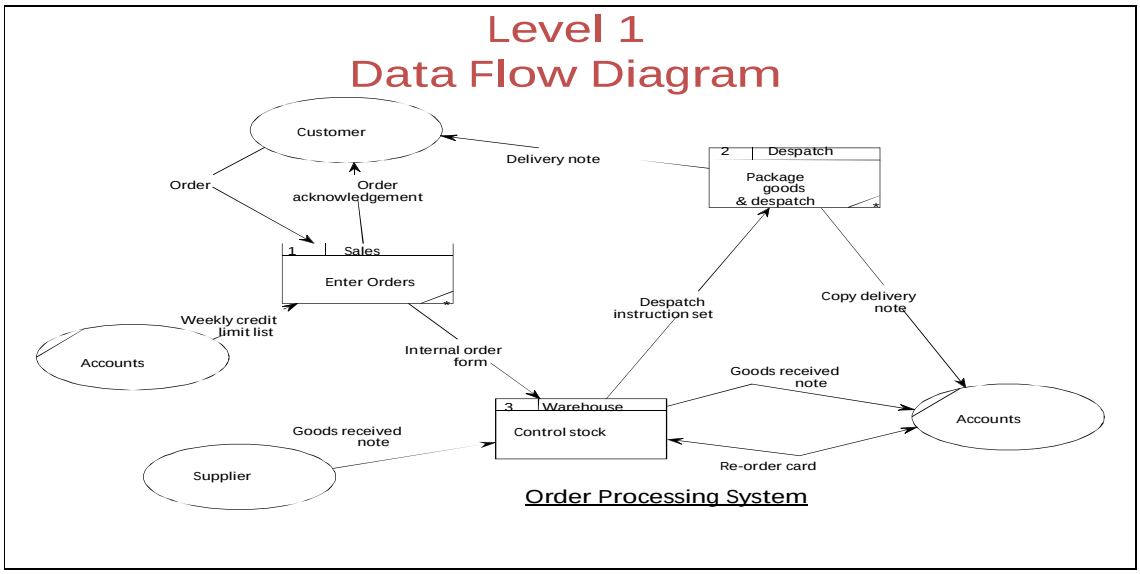
5.3.4 Process Techniques

Process models reveal how the system being developed is used in broader business processes. Data flow models may be used to show the processes and the flow of information from one process to another.

1. Data Flow Diagrams

It provides the key means of achieving one of structured systems. They provide the analyst with the ability to specify a system at the logical level. They represent logical information, not the physical aspects. They have the following characteristics

- Illustrate processing steps in a system
- Tracking and documenting how the data associated with a particular process moves through the system
- Helps analysts and designers understand what is going on
- Simple and intuitive
- Possible to explain them to potential system users who can then participate in validating the model
- Focus on system functions
- Do not recognize system objects (therefore, data flow models not included in UML)
- Similar to activity diagrams in UML 2.0 (data-driven systems are common in business)



2. Elementary process description

Once a DFD is obtained the next step is to precisely specify the process. The methods available for documenting and analyzing the logic of structured decisions include structured English, decision tables, and decision trees. Process specifications are created for primitive processes and some higher level processes on a data flow diagram. They are also are called elementary process diagrams. Structured English, Decision tables and Decision Trees are used to describe processes. Decision tables are used when the process is logically complex involving large number of conditions and alternate solutions. Decision trees are used when conditions to be tested must follow a strict time sequence.

i) Structured English: Structured English is based on structured logic**.** Simple English statements such as add, multiply, move, and so on**.** It is an appropriate technique for analyzing the system when structured decisions are not complex**.** Structured English is similar to a programming language. It does not have strict syntax rules as in programming languages as the intention is only to give precise description of a process. The structured English description should be understandable to the user. The following steps are needed:

- Express all logic in terms of sequential structures, decision structures, case structures, or iterations
- Use and capitalize accepted keywords such as IF, THEN, ELSE, DO, and PERFORM
- Indent blocks of statements to show their hierarchy (nesting) clearly
- Underline words or phrases used have been defined in a data dictionary to signify that they have a specialized, reserved meaning
- Be careful when using "and" and "or"
- Avoid confusion when using logical comparisons such as "greater than" and "greater than or equal to"

ii) Decision Table: Decision tables provide a way to examine, describe, and document decisions using a table. They are used to:

1. Describe the conditions
2. Identify possible decision alternatives
3. Indicate actions should be performed, and describe actions

Decision tables help analysts ensure completeness and accuracy. The four main problems that can occur in developing decision tables are: Incompleteness, Impossible situations, Contradictions and Redundancy

iii) **Decision trees:** Decision trees are used when complex branching occurs in a structured decision process. Trees are also useful when it is essential to keep a string of decisions in a particular sequence. To draw decision trees:

- First, identify all conditions and actions and the order and timing of these (if they are critical)
- Second, begin building the tree from left to right while making sure you are complete in listing all possible alternatives before moving over to the right.

5.3.4.2 Guidelines for choosing EPD

- Use structured English when there are many repetitious actions or when communication to end users is important
- Use decision tables when complex combination of conditions, actions, and rules are found or you require a method that effectively avoids impossible situations, redundancies, and contradictions
- Use decision trees when the sequence of conditions and actions is critical or when not every condition is relevant to every action (the branches are different)

5.3.5Object Oriented Techniques

In the object-oriented design approach, the system is viewed as collection of objects. The state is decentralized among the objects and each object manages its own state information. The functions defined for one object cannot refer or change data of other objects. Objects have their own internal data which define their state. Similar objects constitute a class. In other words, each object is a member of some class. Classes may inherit features from super class. Conceptually, objects communicate by message passing.

5.3.5.1 Origin of UML

In the late 1980s and early 1990s, there was a proliferation of object-oriented design techniques and notations. Different software development houses were using different notations to document their object-oriented designs. These diverse notations gave rise to a lot of confusion. UML was developed to standardize the large number of object-oriented modelling notations that existed and were used extensively in the early 1990s.

UML,(Unified Modelling Language) as the name implies, is a modeling language. It may be used to visualize, specify, construct, and document the artifacts of a software system. It provides a set of notations to create a visual model of the system. Like any other language, UML has its

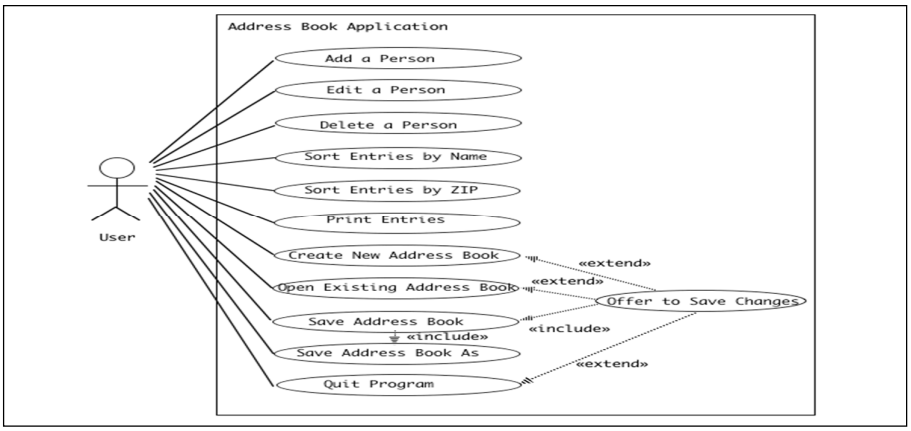
own syntax and semantics .Also, we should clearly understand that UML is not a system design or development methodology, but can be used to document object-oriented and analysis results obtained using some methodology.

1.Use case modeling

Use cases were developed originally to support requirements elicitation and now incorporated into the UML. They are used to model interactions between a system and external actors (users, other systems). It gives a fairly simple overview of an interaction and is used to support requirements elicitation. However you have to provide more detail to understand what is involved: textual description, structured description in a table, sequence diagram. Each use case represents a discrete task that involves external interaction with a system. Actors in a use case may be people or other systems. It is represented diagramatically to provide an overview of the use case and in a more detailed textual form.

Use case:

- Represents a discrete task (involves external interaction with a system)
- Shown as an ellipse with the actors involved in the use case
- Key elements often involved in use cases: actors, description, data, stimulus, response, comments

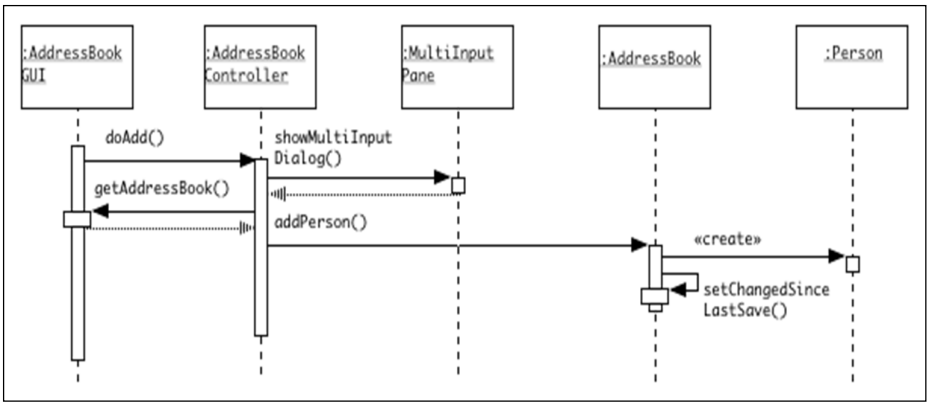


2 Sequence diagrams

Sequence diagrams are part of the UML and are used to model the interactions between the actors and the objects within a system. A sequence diagram shows the sequence of interactions that take place during a particular use case or use case instance.

- Objects and actors involved are listed along the top of the diagram
- Interactions between objects are indicated by annotated arrows
- Rectangle on the dotted lines indicates the lifeline of the object concerned

- Read the sequence of interactions from top to bottom
- Annotations on the arrows indicate the calls to the objects, their parameters and the return values
- Include every interaction in sequence diagrams for code generation or detailed documentation

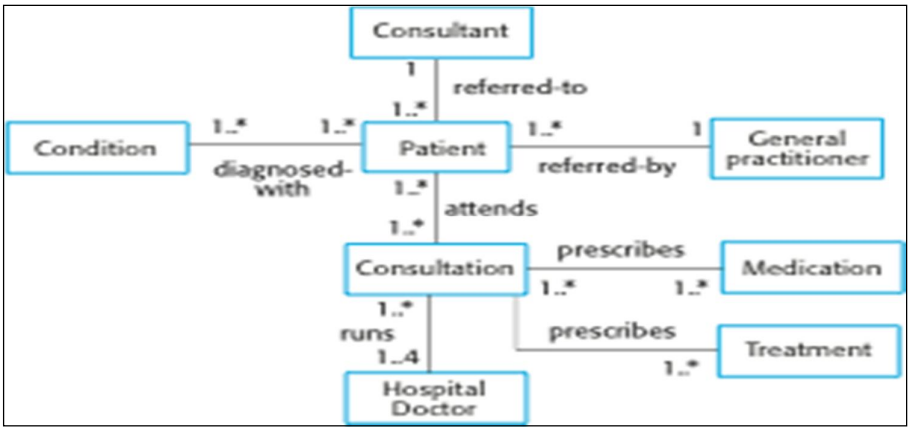


3. Structural models

Structural models of software display the organization of a system in terms of the components that make up that system and their relationships. The models may be static models, which show the structure of the system design, or dynamic models, which show the organization of the system when it is executing. You create structural models of a system when you are discussing and designing the system architecture.

4. Class diagrams

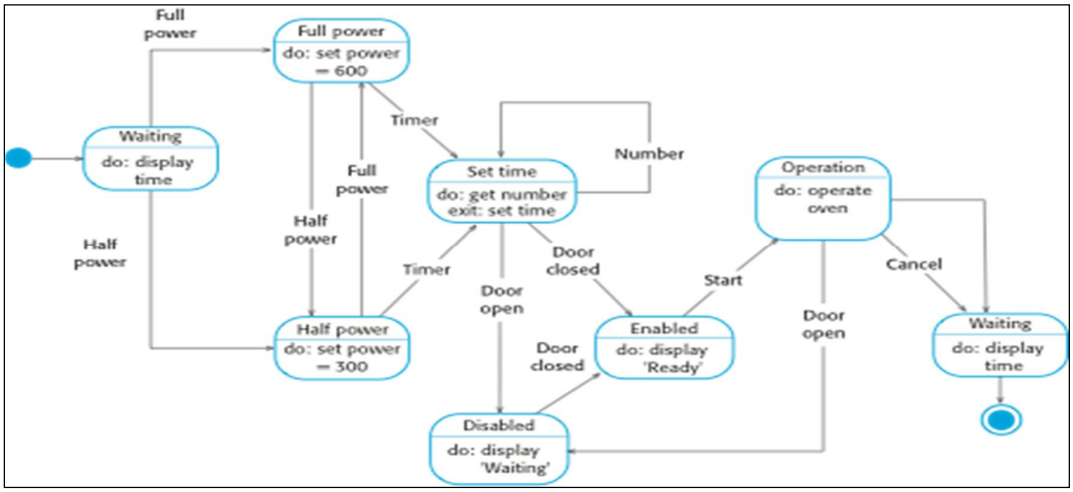
Class diagrams are used when developing an object-oriented system model to show the classes in a system and the associations between these classes. An object class can be thought of as a general definition of one kind of system object. An association is a link between classes that indicates that there is some relationship between these classes. Each class may have some knowledge of its associated class. They are expressed at different levels of detail. The types of relationship: 1 to 1, m to 1, m to n.



5. **State diagrams (UML):**

   They show system states and events that cause transitions from one state to another and o not show the flow of data within the system. It may include additional information on the computations carried out in each state .

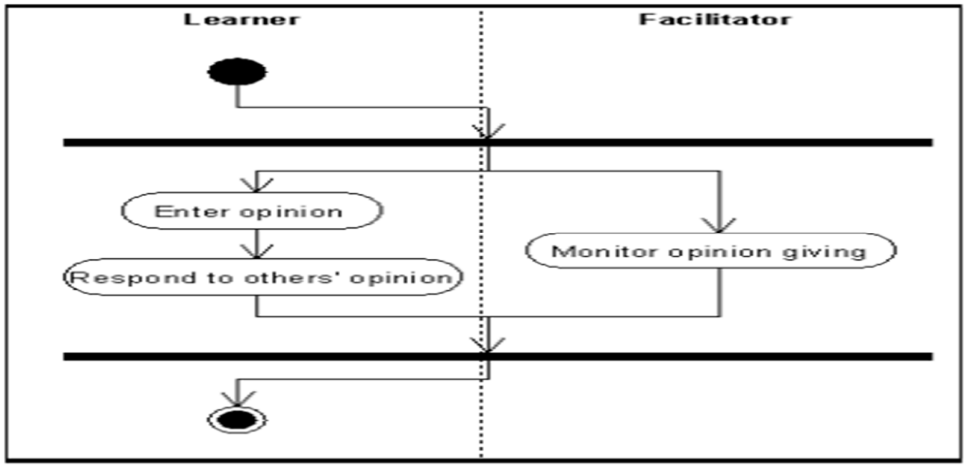
   - Rounded rectangles: represent system states (may include a brief description of the actions taken in that state).
   - Labeled arrows: represent stimuli that force a transition
   - Optional: start and end states using filled circles (see activity diagrams)



6. **UML activity diagrams: They** show the activities that make up a system process and the flow of control from one activity to another .

   - Start of process: indicated by a filled circle
   - End of process: filled circle inside another circle
   - Activities: rectangles with round corners

- Arrows represent the flow of work from one activity to another
- Arrows annotated with guards: indicate the condition when that flow is taken
- Solid bar: indicate activity coordination



5.3.6 Project Management Techniques

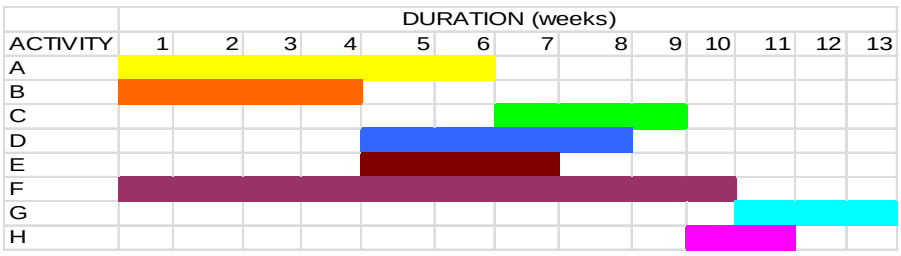
Project Management techniques describe the ways that we gather information, communicate, and generally get things done in the most efficient and effective way.

1.Scheduling Techniques

Project scheduling is concerned with attaching a timescale and sequence to the activities to be conducted within the project. Materials and people needed at each stage of the project are determined and the time each is to take will be set. Project managers have a variety of techniques to develop a project schedule - from the relatively simple process of action planning for small projects, to use of Gantt Charts and Network

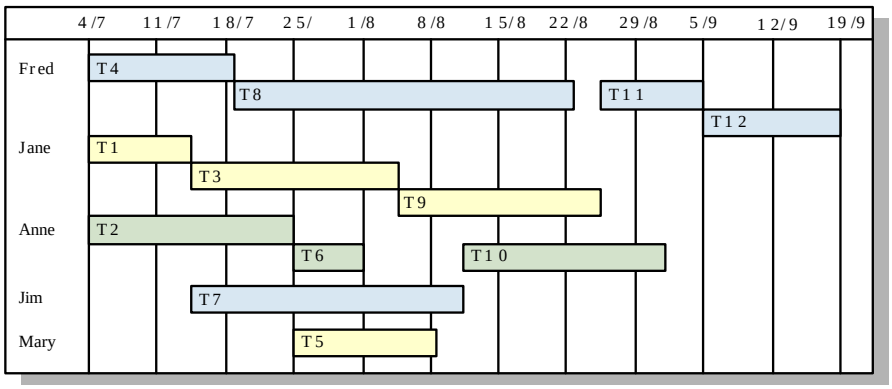
2.Gantt Chart

A Gantt chart, commonly used in project management, is one of the most popular and useful ways of showing activities (tasks or events) displayed against time. On the left of the chart is a list of the activities and along the top is a suitable time scale. Each activity is represented by a bar; the position and length of the bar reflects the start date, duration and end date of the activity.



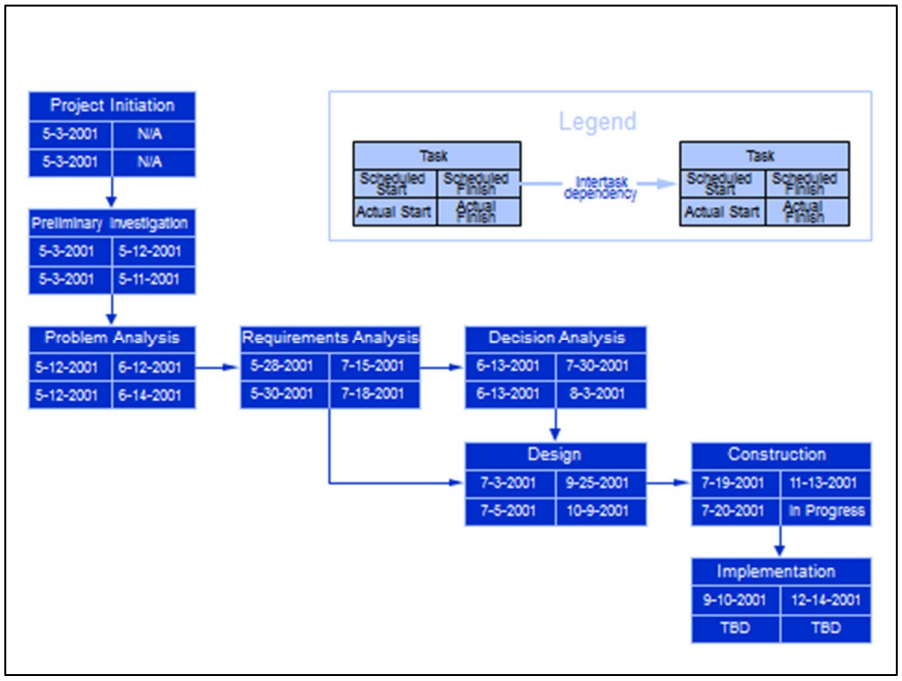
3.Resource Allocation

This is used to allocate resources in the project. The bar chart shows the resource allocation over time.



4. PERT Diagram

A PERT chart is a project management tool used to schedule, organize, and coordinate tasks within a project. PERT stands for *Program Evaluation Review Technique*, A PERT chart presents a graphic illustration of a project as a network diagram consisting of numbered *nodes* (either circles or rectangles) representing events, or milestones in the project linked by labelled *vectors* (directional lines) representing tasks in the project. The direction of the arrows on the lines indicates the sequence of tasks.



5.4 Activities

Read the following topics from the book **Information Systems Development: Methodologies, Techniques and Tools by Avison and Fitzgerald** and answer the questions below

Topics

1. Organisational Techniques
2. People Techniques
3. Techniques in context

Questions

1. Describe the term lateral thinking
2. Distinguish between JAD and Stakeholder analysis
3. Describe the potential benefits and the potential blocks to problem cognition when using techniques in system development

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5.5 Summary

- There are various types of techniques that may be used during system development. These techniques can be classified into different categories. Each technique uses its own notations and presents a different view or perspective of that system.
- Holistic techniques are based upon systems theory, which provides an antidote to conventional, 'reductionist' scientific enquiry – with its tendency to 'reduce' phenomena into smaller and smaller components in order to study and understand them.
- Data-driven models show the sequence of actions involved in processing input data and generating an associated output. They are particularly useful during the analysis of requirements as they can be used to show end-to-end processing in a system
- Process models reveal how the system being developed is used in broader business processes. Data flow models may be used to show the processes and the flow of information from one process to another
- In the object-oriented design approach, the system is viewed as collection of objects. The state is decentralized among the objects and each object manages its own state information.
- Project Management techniques describe the ways that we gather information, communicate, and generally get things done in the most efficient and effective way.

Suggestion for further reading

Extracted from

1. Information Systems Development: Methodologies**,** Techniques and Tools by Avison and Fitzgerald

2. Soft Systems Methodology by Jeremy Rose

3. Project Management techniques by Margret Rouse

4. Systems Analysis and Design by Kendal and Kendal

5. http://www.studytonight.com/dbms/database-normalization.php